INTERFACING DHT11 SENSOR WITH ESP8266 TO MEASURE TEMPERATURE

Mini Project

Subject Code: EC-681

Electronics and Communication Engineering

By

Snay Rai - 25500323070 - Reg No: 232550120443
Barsha Maity - 25500323067 - Reg No:232550120440
Susmita Bhaskar - 25500323072 - Reg No:232550120445
Indrajit Bhowmick - 25500322044 - Reg No:222550110270
Sushovan Singha - 25500322052 - Reg No:222550110294
Sushovan Chandra - 25500323071 -Reg No:232550120444

DR. SUDHIR CHANDRA SUR INSTITUTE OF TECHNOLOGY & SPORTS COMPLEX

Affiliated to Maulana Abul Kalam Azad University (Formerly WBUT)

540 Dum Dum Road, Surer math, W: B: Kolkata-700074,

TABLE OF CONTENT

CONTENT

- **INTRODUCTION:.....**
- **OBJECTIVES:.....**
- **MATERIALS REQUIRED:.....**
- **THEORY:.....**
- **CIRCUIT DIAGRAM:.....**
- **SOFTWARE SETUP: .....**
- **MATERIALS REQUIRED: .....**
- **PROGRAMMING:.....**
- **OUT PUT: .....**
- **REFERENCES:.....**

Introduction

Interfacing the DHT11 Sensor with ESP8266 to Measure Temperature:
The DHT11 is a popular temperature and humidity sensor that can be easily interfaced with microcontrollers like the ESP8266. By connecting the DHT11 sensor to the ESP8266, we can accurately measure temperature readings and send them to a web server or display them on a serial monitor for real-time monitoring. This project demonstrates how to utilize the sensor with the ESP8266 to capture and process temperature data for various applications.

OBJECTIVES:

1. To interface a DHT11 sensor with the ESP8266 NodeMCU board.
2. To measure temperature using the DHT11 sensor.
3. To display the temperature readings on the Blynk IoT app.

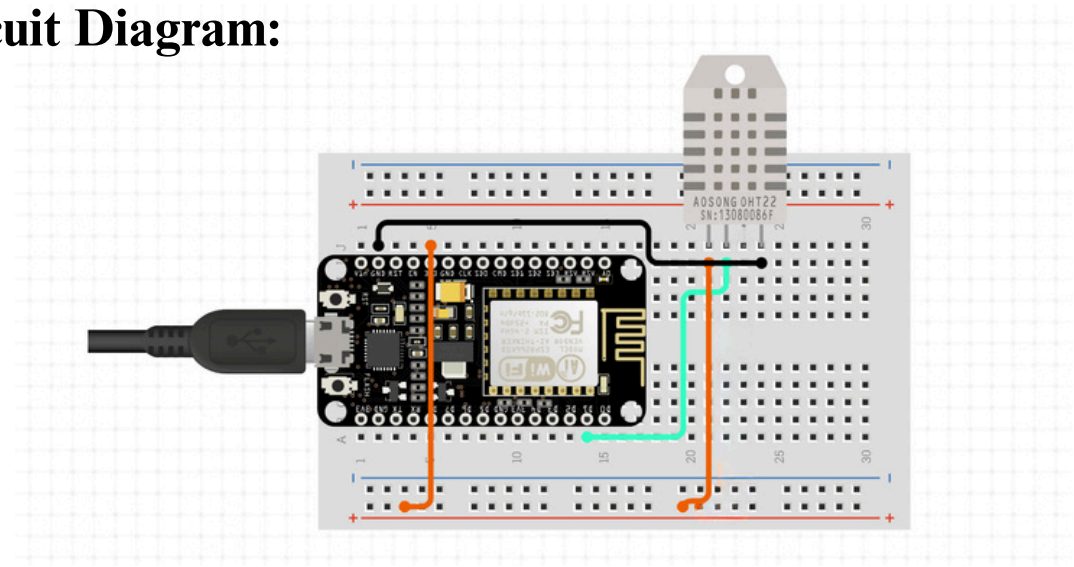
Materials Required:

- ESP8266 NodeMCU board
- DHT11 sensor module
- Breadboard
- Jumper wires
- USB cable
- Computer with Arduino IDE installed
- Blynk app installed on a smartphone or tablet

Theory:

The DHT11 sensor is a basic, low-cost digital temperature sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and spits out a digital signal on the data pin. It is quite simple to use but requires careful timing to grab data. The ESP8266 NodeMCU board is a low-cost Wi-Fi microchip with full TCP/IP stack and microcontroller capability, designed to provide access to Wi-Fi networks or to act as an access point. Blynk is a platform with iOS control Arduino, Raspberry Pi, and the likes over the Internet. It's a digital dashboard where you can build a graphic interface for your project by simply

Circuit Diagram:



Software Setup:

- Open the Arduino IDE on your computer.
- Install the necessary libraries: DHT sensor library and Adafruit Unified Sensor library.
- Install the Blynk library by navigating to Sketch
> Include Library
> Manage Libraries and searching for “Blynk”.

Blynk App Setup:

- Open the Blynk app on your smartphone/tablet.
- Create a new project and note the Auth Token.
- Add two Value Display widgets and set them to Virtual Pin V5 and V6, respectively.

Materials Required:

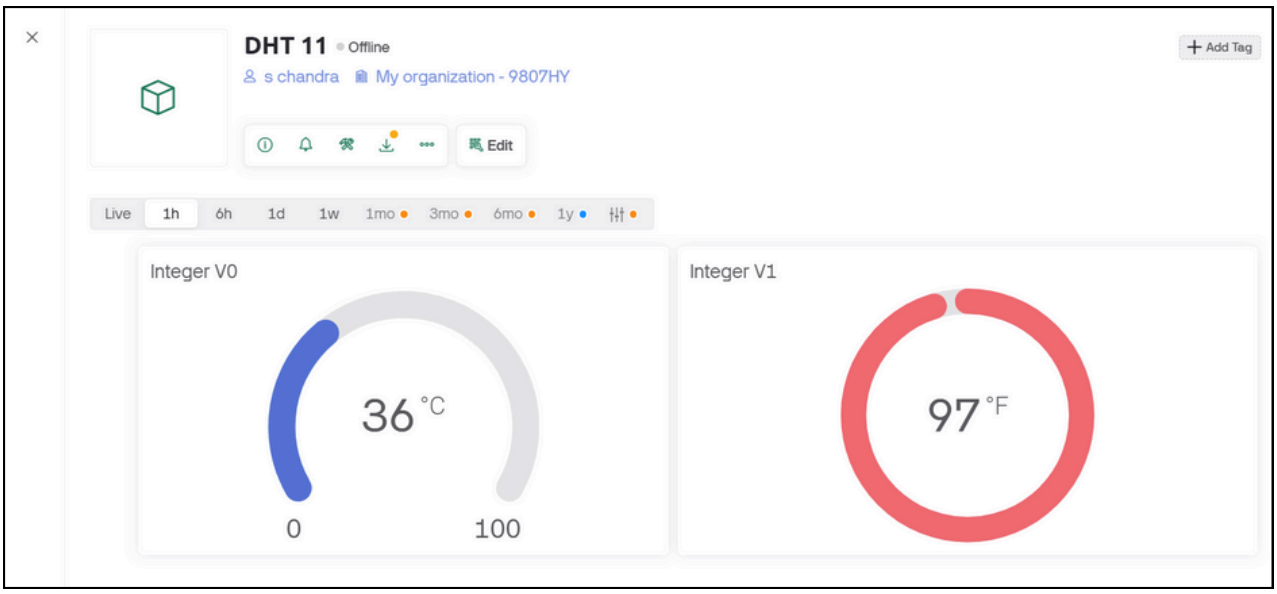
- ESP8266 NodeMCU board
- DHT11 sensor module
- Breadboard
- Jumper wires
- USB cable
- Computer with Arduino IDE installed
- Blynk app installed on a smartphone or tablet

Programming:

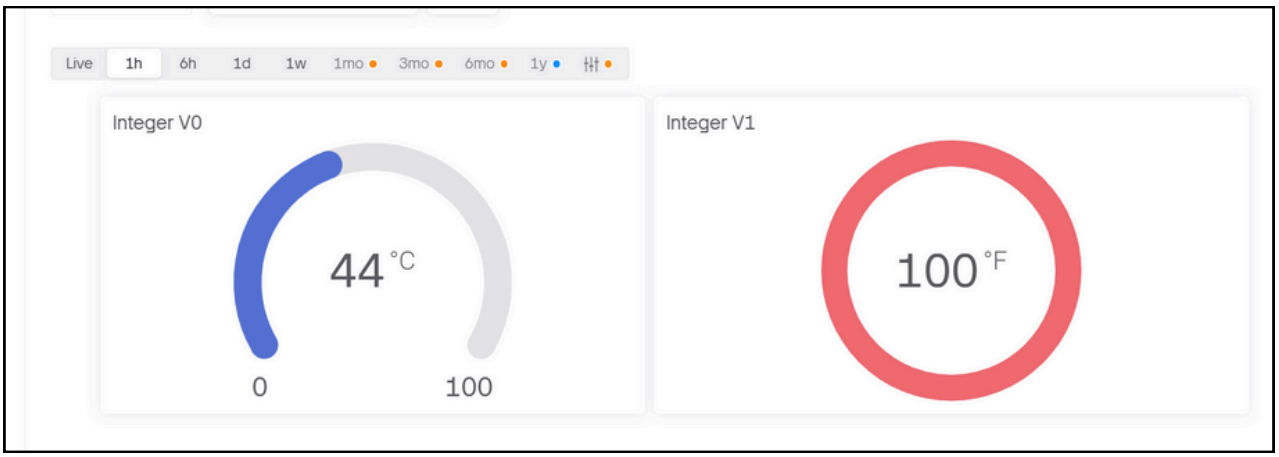
```
#define BLYNK_TEMPLATE_ID "TMPL3Y49g1Kjb"
#define BLYNK_TEMPLATE_NAME "DHT 11"
#define BLYNK_AUTH_TOKEN "QxxuyhbcQQoMgnFvBGoI15SggoR"

#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>

// Your network credentials
char ssid[] = "Ahead"; //your wifi name//
char pass[] = "20242024"; //your wifi password//

// Blynk template details


// DHT sensor settings
#define DHTPIN 5 // Pin where the DHT11 is connected
#define DHTTYPE DHT11 // DHT 11
DHT dht(DHTPIN, DHTTYPE);

// Timer settings
BlynkTimer timer;

void sendSensor() {
 float tC = dht.readTemperature(); // Read temperature in
Celsius
 float tF = dht.readTemperature(true); // Read temperature in
Fahrenheit

 if (isnan(tC) || isnan(tF)) {
  Serial.println("Failed to read from DHT sensor!");
  return;
 }
 Blynk.virtualWrite(V0, tC); // Temperature in Celsius to
virtual pin V5 (Gauge)
 Blynk.virtualWrite(V1, tF); // Temperature in Fahrenheit to
virtual pin V6 (Label)
}

void setup() {
 Serial.begin(9600);
 Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
 dht.begin();

 // Setup a function to be called every 2 seconds
 timer.setInterval(2000L, sendSensor);
}

void loop() {
 Blynk.run();
 timer.run();
}
```

Out put:

Before Running:



After Running:



Running Video :

REFERENCES

1. P.chakravarth, S.Natarajan, M.Anto Bennete "GSM based soldier tracking system and monitoring using wireless communication" Department of Electronics and Communication, published in September 1st ,2017.
2. Akshita v.Armarkar, Deepika J, Punekar, Mrunali, p.Kapse, Shweta Kumari, Jayshree Shelka "Soldier health and position tracking system" Department of ETC engineering , volume 7 issue no..3
3. Patil Akshay , Shelake Balaji, Pinjari Raju, Mirajkar P.P "GPS based soldier tracking and health monitoring", Department of ETC engineering, volume 40 issue:03 March 2017.
4. Shruthi Nikam, Supriya Patil, Prajkata Powar, V.S.Bendre "GPS based soldier tracking and health indication system", Department of E&TC PIMPRI CHINCHWA COLLEGE OF
5. Govindiraj A, Dr.S.Sindhurja Banu "GPS based soldier tracking and health indication system with environmental analysis", Department of Electronics and Communicaton Engineering volume 2, issue 12, Decemember 2013.
6. Zeeshan Raza,Kamaran Liaquat,Shahazad Ashraf "Monitoring of soldier health and transmition of secret codes",NFC instutation of engineering volume 8 no 2,june 2017.
7. Shubhangi Gupta, Shivani Kulshrestha, Divya Singh, Ashish Kumar, Er.Hitendra Singh "GPS and GSM based soldier health monitoring and tracking system" issue 3 volume 6, March 2017. r81 JouniRantakoko,
8. JoakimRydell and peter Sromback, "Accurate and Realiable soldier and first responder Poasitioning :Multisensor System and co-operative localization"' April-2011.
9. RavidraB.Sathe ⊠A.S.Bhide. "Gps-based soldier tracking and health monitoring system" conference on Advances in communication and computing April 2011-12.
10. VincentPereira,Audrey Giremus,andericgrivel "Modeling of multipath environment using copulas For particle filtering based GPS Navigation" June-2012. 11 ENGINEERING, volume 2, issue 3, March 2013.